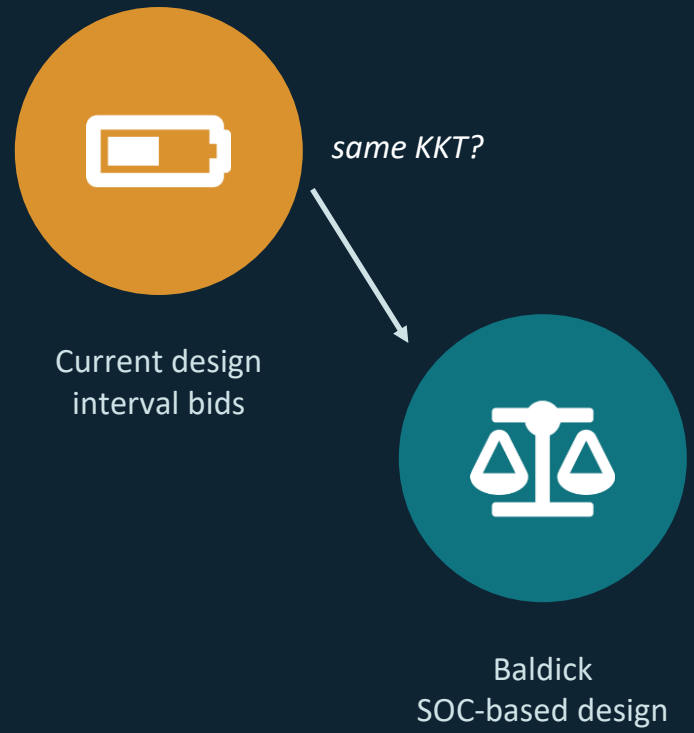
THESIS PROGRESS · ESS IN DAY-AHEAD MARKETS

Can current-market bids reproduce Baldick's SOC-based day-ahead outcome?

KKT equivalence between interval-by-interval ESS bids and the SOC-based design, and the experimental plan



Carlos · September 2026

MOTIVATION

The question

Current design (CD): ERCOT DAM

Each ESS submits interval-by-interval charge bids and discharge offers. SOC is not in the clearing; the ESS manages it through its bids.

SOC-based design (Baldick, 2026)

The ISO optimizes SOC. The ESS submits only technical data and one end-of-horizon bid b_e. Benchmark for social welfare.

If the regulator keeps the current design, do there exist charge bids and discharge offers that reproduce Baldick's outcome?

Same KKT conditions ⇒ same dispatch and same nodal prices.

What information must those bids embed, and can an ESS build them without Baldick's solution?

Approach

1. Write both clearing problems and their KKT conditions in full
2. Match the stationarity conditions term by term
3. Read off the equivalent bids and the information they require
4. Quantify the welfare gap when that information is missing

Steps 1–3: done (equivalence note) · Step 4: experimental plan

Notation at a glance

Sets	
$n \in N,\ \ell \in L,\ r$	buses, lines, reference bus
$i \in G,\ k \in K_i$	generating units, offer blocks
$m \in D,\ j \in J_m$	loads, bid blocks
$e \in E,\ t \in T$	storage units, intervals (length ΔT)
Variables	
$q_{i,k,t},\ d_{m,j,t}$	generation, served elastic demand (MW)
$f_{\ell,t},\ \theta_{n,t}$	line flow (MW), voltage angle
$p_{e,t},\ g_{e,t},\ s_{e,t}$	ESS charge, discharge (MW); SOC (MWh)

ESS data	
$\alpha_e,\ \beta_e,\ \gamma_e$	charge / inverse-discharge coeff.; retention
$\bar{s}_e, \underline{s}_e, \bar{p}_e, \bar{g}_e$	SOC and power limits
$\varepsilon_e,\ \zeta_e$	degradation cost, charge / discharge
$b_e(\cdot),\ w_e(\cdot)$	terminal bid benefit and its slope (SOC design)
$u^p{}_{e,j,t},\ c^g{}_{e,k,t};\ \hat{p}, \hat{g}$	CD charge-bid, discharge-offer prices; block sizes
Multipliers ($/MWh)	
$\lambda_{n,t},\ \pi_{e,t}$	nodal price (LMP); marginal value of SOC
$\bar{\nu}_{e,t}, \underline{\nu}_{e,t};\ \rho, \sigma$	SOC-limit rents; charge / discharge bounds

Multipliers of MW constraints enter the Lagrangian with factor ΔT, so every multiplier is in $/MWh. (·)★ denotes the optimum of the SOC design.

The rest of the market is identical in both designs

Constraints (multiplier)

$$\sum_{m\in\mathcal{D}(n),j} d_{m,j,t} + D_{n,t} + \sum_{\ell} A_{n\ell} f_{\ell,t} - \sum_{i\in\mathcal{G}(n),k} q_{i,k,t} - \sum_{e\in\mathcal{E}(n)} \left(g_{e,t} - p_{e,t}\right) = 0 \qquad \lambda_{n,t} \qquad \text{(A1)}$$

$$f_{\ell,t} - y_\ell \sum_n A_{n\ell}\theta_{n,t} = 0 \qquad \kappa_{\ell,t} \qquad \text{(A2)}$$

$$-\bar{f}_\ell \le f_{\ell,t} \le \bar{f}_\ell \qquad \underline{\mu}_{\ell,t}, \bar{\mu}_{\ell,t} \qquad \text{(A3)}$$

$$\theta_{r,t} = 0 \qquad \xi_t \qquad \text{(A4)}$$

$$0 \le q_{i,k,t} \le \bar{q}_{i,k,t} \qquad \underline{\delta}_{i,k,t}, \bar{\delta}_{i,k,t} \qquad \text{(A5)}$$

$$Q_{i,t} - Q_{i,t-1} \le \bar{R}_i, \quad Q_{i,t-1} - Q_{i,t} \le \underline{R}_i, \quad t \ge 2 \qquad \bar{\tau}_{i,t}, \underline{\tau}_{i,t} \qquad \text{(A6)}$$

$$0 \le d_{m,j,t} \le \bar{d}_{m,j} \qquad \underline{\psi}_{m,j,t}, \bar{\psi}_{m,j,t} \qquad \text{(A7)}$$

Stationarity (S1)–(S4)

$$\partial q_{i,k,t}: \quad c_{i,k} - \lambda_{n(i),t} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + \varrho_{i,t} = 0 \qquad \text{(S1)}$$

$$\partial d_{m,j,t}: \quad -u_{m,j} + \lambda_{n(m),t} + \bar{\psi}_{m,j,t} - \underline{\psi}_{m,j,t} = 0 \qquad \text{(S2)}$$

$$\partial f_{\ell,t}: \quad \sum_n A_{n\ell}\lambda_{n,t} + \kappa_{\ell,t} + \bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t} = 0 \qquad \text{(S3)}$$

$$\partial \theta_{n,t}: \quad -\sum_\ell y_\ell A_{n\ell}\kappa_{\ell,t} + \xi_t \mathbf{1}_{\{n=r\}} = 0 \qquad \text{(S4)}$$

$$\varrho_{i,t} := \left(\bar{\tau}_{i,t} - \underline{\tau}_{i,t}\right) - \left(\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1}\right)$$

- DC network with nodal prices $\lambda_{n,t}$; the ESS sees the price at its own bus n(e).
- $\varrho_{i,t}$ is the ramp wedge; ramps apply for t ≥ 2.
- Complementary slackness (CS1)–(CS3) for all bounds, omitted here.

Current design: interval bids, no SOC in the clearing

Clearing problem

$$\min \; f^{\mathrm{CD}} = \sum_t \Delta T \Big[\sum_{i,k} c_{i,k} q_{i,k,t} - \sum_{m,j} u_{m,j} d_{m,j,t} + \sum_e \Big(\sum_k c^g_{e,k,t}\, g_{e,k,t} - \sum_j u^p_{e,j,t}\, p_{e,j,t} \Big) \Big]$$

$$(\mathrm{A1})\text{–}(\mathrm{A7}), \qquad g_{e,t} = \textstyle\sum_k g_{e,k,t}, \quad p_{e,t} = \sum_j p_{e,j,t}$$

$$0 \le g_{e,k,t} \le \hat{g}_{e,k,t}, \quad \textstyle\sum_k \hat{g}_{e,k,t} \le \bar{g}_e \qquad \underline{\sigma}_{e,k,t}, \bar{\sigma}_{e,k,t} \quad (\mathrm{C1})$$

$$0 \le p_{e,j,t} \le \hat{p}_{e,j,t}, \quad \textstyle\sum_j \hat{p}_{e,j,t} \le \bar{p}_e \qquad \underline{\rho}_{e,j,t}, \bar{\rho}_{e,j,t} \quad (\mathrm{C2})$$

KKT: ESS rows (the rest is (S1)–(S4))

$$\partial g_{e,k,t}: \quad c^g_{e,k,t} - \lambda_{n(e),t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t} = 0 \qquad (\mathrm{S5^{CD}})$$

$$\partial p_{e,j,t}: \quad -u^p_{e,j,t} + \lambda_{n(e),t} + \bar{\rho}_{e,j,t} - \underline{\rho}_{e,j,t} = 0 \qquad (\mathrm{S6^{CD}})$$

Box form

$$0 \le g_{e,k,t} \le \hat{g}_{e,k,t} \;\perp\; c^g_{e,k,t} - \lambda_{n(e),t}$$

$$0 \le p_{e,j,t} \le \hat{p}_{e,j,t} \;\perp\; \lambda_{n(e),t} - u^p_{e,j,t}$$

- ESS = paired generator (offer blocks c^g) + load (bid blocks u^p).
- Stepped bids per interval, as in ERCOT.
- SOC is not represented: the ESS must build its intertemporal plan into its bids.
- ERCOT DAM: nodal (A1)–(A4): collapse the network to one zone.
- Simplified: no unit commitment, AS or virtual bids.

Baldick's design: the ISO optimizes SOC

Clearing problem

$$\min \; f^{\mathrm{SOC}} = \sum_t \Delta T \Big[\sum_{i,k} c_{i,k} q_{i,k,t} - \sum_{m,j} u_{m,j} d_{m,j,t} + \sum_e \big(\epsilon_e p_{e,t} + \zeta_e g_{e,t} \big) \Big] - \sum_e b_e \big(s_{e,T} - \bar{\bar{s}}_{e,T} \big)$$

subject to

$$(\mathrm{A1})\text{–}(\mathrm{A7})$$

$$s_{e,t} - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} + \Delta T \beta_e g_{e,t} = 0 \qquad \pi_{e,t} \qquad (\mathrm{A8})$$

$$\gamma_e s_{e,t-1} + \Delta T \alpha_e p_{e,t} \le \bar{s}_e \qquad \bar{\nu}_{e,t} \qquad (\mathrm{A9})$$

$$\gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} \ge \underline{s}_e \qquad \underline{\nu}_{e,t} \qquad (\mathrm{A10})$$

$$0 \le p_{e,t} \le \bar{p}_e \qquad \underline{\rho}_{e,t}, \bar{\rho}_{e,t} \qquad (\mathrm{A11})$$

$$0 \le g_{e,t} \le \bar{g}_e \qquad \underline{\sigma}_{e,t}, \bar{\sigma}_{e,t} \qquad (\mathrm{A12})$$

Baldick (2026): (A8) = eq. (4); (A9)–(A10) = eqs. (6)–(7); (A11)–(A12) = eqs. (1)–(2)

- The ESS submits no interval bids, only technical data and the terminal bid b_e.
- $\bar{\bar{s}}_{e,T}$ is the end SOC the ESS would reach without participating.
- (A9)–(A10) are Baldick's tightened SOC limits.
- Exclusivity p ⊥ g relaxed, as in Baldick's examples.

R. Baldick, "A state-of-charge based formulation for storage participation in electricity markets," arXiv:2606.14993, 2026.

KKT conditions of Baldick's design: ESS rows

$$\partial p_{e,t}: \quad \epsilon_e + \lambda_{n(e),t} - \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big) + \bar{\rho}_{e,t} - \underline{\rho}_{e,t} = 0 \tag{S5}$$

$$\partial g_{e,t}: \quad \zeta_e - \lambda_{n(e),t} + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t} = 0 \tag{S6}$$

$$\partial s_{e,t},\ t < T: \quad \pi_{e,t} = \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big) \tag{S7}$$

$$\partial s_{e,T}: \quad \pi_{e,T} \in \partial b_e\big(s_{e,T} - \bar{\bar{s}}_{e,T}\big) \quad \big(= w_e(\cdot) \text{ if differentiable}\big) \tag{S8}$$

+ (S1)–(S4) for the rest of the market, and complementary slackness for power bounds (CS4) and SOC limits (CS5).

Unrolled costate, from (S7)–(S8)

$$\pi_{e,t} = \gamma_e^{T-t}\,\pi_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}\big(\underline{\nu}_{e,\tau} - \bar{\nu}_{e,\tau}\big)$$

- $\pi_{e,t}$ is the value of one more MWh stored at the end of t.
- It equals the discounted terminal value plus the future SOC-limit rents.
- It is constant between SOC-binding events ($\gamma_e = 1$).

KKT side by side: only the ESS rows differ

Condition	Current design	SOC design (Baldick)
Rest of market (S1)–(S4), (CS1)–(CS3), (A1)–(A7)	*identical*	*identical*
Charge stationarity	$\lambda_{n(e),t} = u^{p}_{e,j,t} - \bar{\rho}_{e,j,t} + \underline{\rho}_{e,j,t}$	$\lambda_{n(e),t} = \left[\alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e\right] - \bar{\rho}_{e,t} + \underline{\rho}_{e,t}$
Discharge stationarity	$\lambda_{n(e),t} = c^{g}_{e,k,t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t}$	$\lambda_{n(e),t} = \left[\zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t})\right] + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t}$
ESS power bounds	$\hat{p}_{e,j,t},\ \hat{g}_{e,k,t}$ (bid quantities)	$\bar{p}_e,\ \bar{g}_e$ (physical limits)
SOC rows	*absent*	(A8)–(A10), (S7)–(S8)

Reading: the CD bid prices (amber) play exactly the role of the bracketed terms of the SOC design (teal). Matching them makes the two KKT systems coincide.

Equivalent bids and why they work

(E)

$$u^{p\star}_{e,t} = \alpha_e\left(\pi^\star_{e,t} - \bar{\nu}^\star_{e,t}\right) - \epsilon_e, \quad \hat{p}_{e,t} = \bar{p}_e \qquad c^{g\star}_{e,t} = \zeta_e + \beta_e\left(\pi^\star_{e,t} + \underline{\nu}^\star_{e,t}\right), \quad \hat{g}_{e,t} = \bar{g}_e$$

Proposition 1: KKT equivalence

Let $z^\star$ be primal–dual optimal for Baldick's design. Under bids (E), $z^\star$ satisfies every KKT row of the current design.

⇒ same dispatch and the same nodal prices $\lambda^\star$.

Proof: substitute (E) into (S5)–(S6) to get ($S5^{CD}$)–($S6^{CD}$) with the same $\rho^\star$, $\sigma^\star$. The bounds coincide ($\hat{p} = \bar{p}$, $\hat{g} = \bar{g}$), the rest is common, and the CD is an LP, so KKT is sufficient.

Proposition 2: CD is a Lagrangian relaxation

Dualize the SOC rows (A8)–(A10) of Baldick's problem at $(\pi^\star, \bar{\nu}^\star, \underline{\nu}^\star)$:

$$\min_s \ \mathcal{L}^{\mathrm{SOC}}\left(\cdot\,; \pi^\star, \bar{\nu}^\star, \underline{\nu}^\star\right) = f^{\mathrm{CD}} + K \qquad \Rightarrow \qquad f^{\mathrm{SOC}\star} = f^{\mathrm{CD}\star} + K$$

$$K = \sum_e \left[-\pi^\star_{e,0} s_{e,0} - \bar{s}_e \sum_t \bar{\nu}^\star_{e,t} + \underline{s}_e \sum_t \underline{\nu}^\star_{e,t} + \min_s \left\{ \pi^\star_{e,T} s - b_e(s - \bar{\bar{s}}_{e,T}) \right\} \right]$$

(ii) A CD optimum is SOC-optimal iff some s satisfies (A8)–(A10) with complementary slackness w.r.t. $\bar{\nu}^\star$, $\underline{\nu}^\star$.

The bids are the SOC design's dual prices.

Stepped bids: when is Baldick's schedule reproduced?

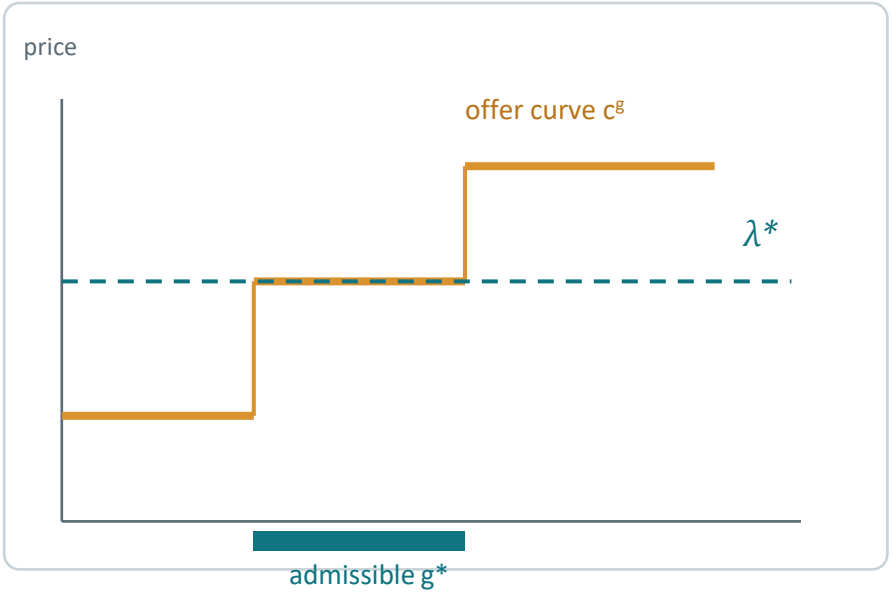
Proposition 3: necessary and sufficient, any stepped bids

$$\sum_{j:\, u^{p}_{e,j,t} > \lambda^\star} \hat{p}_{e,j,t} \;\le\; p^\star_{e,t} \;\le\; \sum_{j:\, u^{p}_{e,j,t} \ge \lambda^\star} \hat{p}_{e,j,t}$$

$$\sum_{k:\, c^{g}_{e,k,t} < \lambda^\star} \hat{g}_{e,k,t} \;\le\; g^\star_{e,t} \;\le\; \sum_{k:\, c^{g}_{e,k,t} \le \lambda^\star} \hat{g}_{e,k,t}$$

$$\lambda^\star := \lambda^\star_{n(e),t}$$

Baldick's schedule must lie on each ESS's submitted bid and offer curves, evaluated at Baldick's nodal price.



Remark: uniqueness and quantities

The CD-optimal set contains the SOC-optimal set but may be larger, possibly SOC-infeasible, where an ESS block is priced at λ★. Tight blocks pin the schedule but may move λ★ where the ESS sets the price. Exact reproduction needs prices and quantities.

Remark: charge/discharge exclusivity

$$c^{g\star}_{e,t} - u^{p\star}_{e,t} = \epsilon_e + \zeta_e + (\beta_e - \alpha_e)\pi^\star_{e,t} + \beta_e \underline{\nu}^\star_{e,t} + \alpha_e \bar{\nu}^\star_{e,t} > 0 \;\Rightarrow\; p_{e,t}\, g_{e,t} = 0$$

With π★ ≥ 0, bids (E) never clear charging and discharging in the same interval.

What information do the equivalent bids embed?

Corollary 1: unrolling π*

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \left[\gamma_e^{T-t}\, \pi^{\star}_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t} \left(\underline{\nu}^{\star}_{e,\tau} - \bar{\nu}^{\star}_{e,\tau} \right) + \underline{\nu}^{\star}_{e,t} \right]$$

$$u^{p\star}_{e,t} = \alpha_e \left[\gamma_e^{T-t}\, \pi^{\star}_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t} \left(\underline{\nu}^{\star}_{e,\tau} - \bar{\nu}^{\star}_{e,\tau} \right) - \bar{\nu}^{\star}_{e,t} \right] - \epsilon_e$$

Private to ESS e

α_e, β_e, γ_e, ε_e, ζ_e and w_e: exactly what e submits in Baldick's design.

Endogenous

Own SOC-limit rents $\bar{v}^{\star}$, $\underline{v}^{\star}$ (and $s^{\star}_{e,T}$ if b_e is nonlinear). They depend on the whole market, but only through e's nodal price path.

Corollary 2: when private information is enough

$$b_e(\Delta) = w_e \Delta, \quad \bar{\nu}^{\star}_{e,\tau} = \underline{\nu}^{\star}_{e,\tau} = 0 \;\; \forall \tau \geq t \quad \Longrightarrow \quad c^{g\star}_{e,t} = \zeta_e + \beta_e \gamma_e^{T-t} w_e, \qquad u^{p\star}_{e,t} = \alpha_e \gamma_e^{T-t} w_e - \epsilon_e$$

Otherwise the ESS needs the rents $v^{\star}$, i.e. a forecast of its intra-horizon nodal prices (Baldick's critique of interval bids).

Two kinds of “current design”, two different gaps

ERCOT DAM

No SOC in the day-ahead clearing

- Equivalent bids need the SOC rents $v^\star$ (Cor. 1).
- The ESS must forecast its nodal price path.
- Cleared schedules can be SOC-infeasible.
- **Gap source: missing SOC rents.**

CAISO · NYISO

SOC in the clearing, interval bids

- The market computes the rents $v^\star$ itself.
- With linear b_e, the private-only bids of Cor. 2 make the problem identical to Baldick’s (Remark 3).
- **Gap source: only a nonlinear terminal bid.**

Why Remark 3 holds: the terminal bid is linear in the schedule

$$-w_e\left(s_{e,T} - \bar{\bar{s}}_{e,T}\right) = \Delta T \sum_t \left[\left(\beta_e \gamma_e^{T-t} w_e\right) g_{e,t} - \left(\alpha_e \gamma_e^{T-t} w_e\right) p_{e,t}\right]$$

⇒ The experiments must report the two current designs separately.

EXPERIMENTAL PLAN

Under perfect information the equivalence is a theorem. The experiments measure the gap when its conditions fail.

How often SOC limits bind

Cor. 2: private bids suffice only if no SOC limit binds

Forecast quality

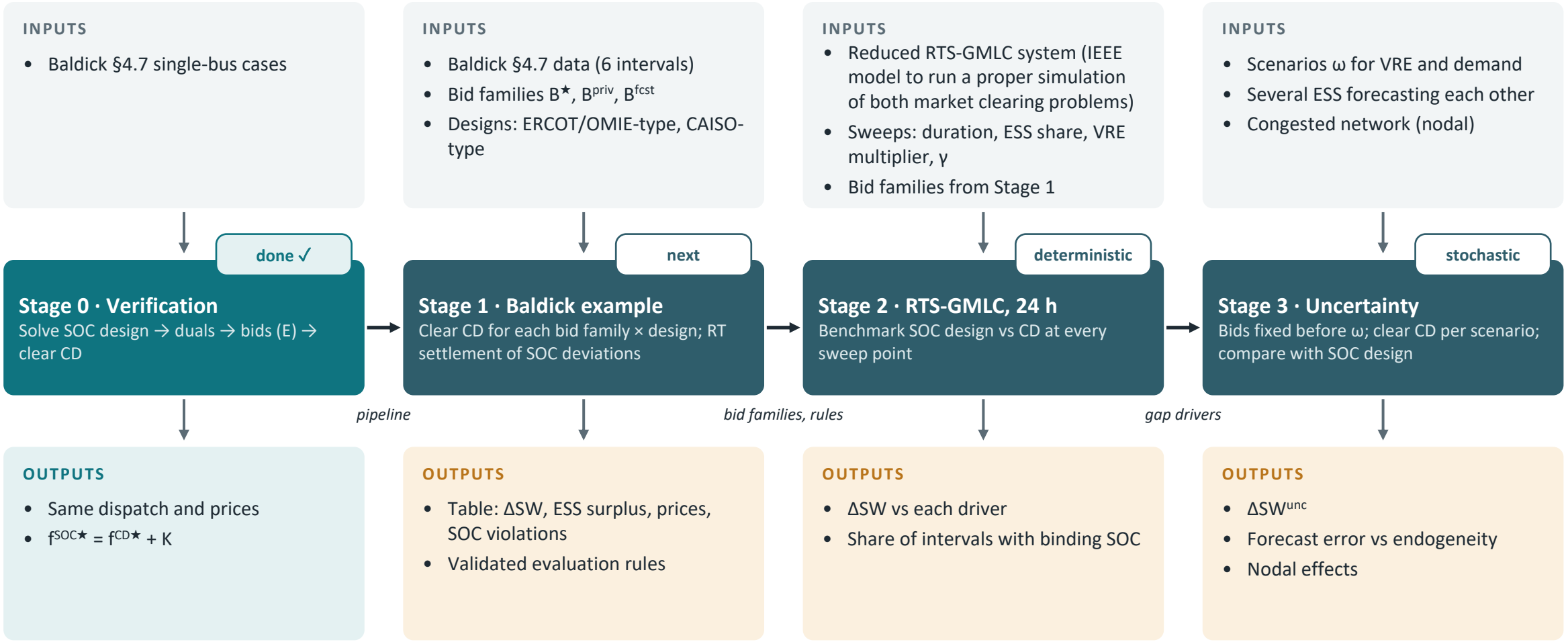
Cor. 1: the missing rents v* come from the nodal price path

Quantity information

Remark 1: prices alone may not pin the schedule

EXPERIMENTAL PLAN

Roadmap: four stages, each feeding the next



Metric throughout: $\Delta SW = SW^{SOC} - SW^{CD}(B)$, with welfare evaluated using the true ESS costs (ε, ζ, w), not the bids.

Stage 1: Baldick's 6-interval example

Data
Baldick §4.7: base, high WTP, $\bar{s} = 1200$

→ **Build bids**
$B^{\star}$, B^{priv}, B^{fcst}

→ **Clear CD**
ERCOT/OMIE-type and CAISO-type

→ **RT settlement**
SOC deviations at realized marginal cost

→ **Outputs**
ΔSW, ESS surplus, prices, SOC violations

Bid family	How it is built	Information used	Expected gap
$B^{\star}$ (sanity check)	(E) with the true $\pi^{\star}$, $\nu^{\star}$	SOC-design duals	0 in both designs
B^{priv}	Cor. 2: $\zeta + \beta\gamma^{T-t}w$, $\alpha\gamma^{T-t}w - \varepsilon$	private data only	> 0 in ERCOT/OMIE-type when SOC binds; 0 in CAISO-type (linear b)
B^{fcst}	Self-schedule on forecast prices → $\hat{v}$ → (E)	private data + price forecast	depends on forecast error

Forecast-based bids

$$\hat{\lambda}_{n(e),t} \longrightarrow \max_{p,g,s} \text{ ESS profit s.t. (A8)–(A12)} \longrightarrow \hat{\pi}_{e,t}, \hat{\bar{\nu}}_{e,t}, \hat{\underline{\nu}}_{e,t} \longrightarrow \text{(E)}$$

BEFORE CODING

Three modeling decisions to fix first

Evaluating SOC-infeasible CD outcomes

- ERCOT DAM is financial: settle deviations in real time at the realized marginal cost.
- ERCOT: model the correction through the intraday markets.
- Without a rule, CD welfare is not physically achievable.

Price selection under degeneracy

- Baldick's example has price ranges (interval 1: 0 to 43.1 $/MWh).
- Welfare is unaffected, but ESS surplus is not.
- Fix one rule (solver duals or minimum price) everywhere.

Scope of the model

- Convex: no unit-commitment binaries.
- Exclusivity $p \perp g$ relaxed, as in Baldick.
- Welfare measured with true ESS costs, not bids.

NEXT STEPS

From equivalence to measured gaps

1. Agree on the two current-design benchmarks: ERCOT-type and Baldicks-type
2. Fix the evaluation and price-selection rules
3. Code Stage 1: three bid families × two designs on Baldick's example
4. Run the Stage 2 sweeps on the reduced RTS-GMLC system

Reference: R. Baldick, "A state-of-charge based formulation for storage participation in electricity markets: Technical reference," arXiv:2606.14993, June 2026.